

RESEARCH ARTICLE

A Lightweight, Real-Time Human Detection and Tracking System for Privacy-Preserving Occupancy Monitoring Using Low-Resolution Thermal Sensors

SUHAS PAPANASHI¹, (Member, IEEE), NISHANT VASANTKUMAR HEGDE², (Member, IEEE),
MANOJITH BHAT V.³, R. BHAGYA¹, (Senior Member, IEEE),
B. ROJA REDDY¹, (Senior Member, IEEE), AND SATISH PATEL⁴, (Senior Member, IEEE)

¹Department of Electronics and Telecommunication Engineering, RV College of Engineering, Bengaluru 560059, India

²Department of Computer Science and Engineering, RV College of Engineering, Bengaluru 560059, India

³Department of Information Science and Engineering, RV College of Engineering, Bengaluru 560059, India

⁴DyuLabs, Ahmedabad, Gujarat 380051, India

Corresponding author: Suhas Papanashi (suhaspapanashi.et22@rvce.edu.in)

ABSTRACT Low-resolution thermal array sensors offer an inherently privacy-preserving alternative to camera-based occupancy monitoring, but their low pixel resolution poses significant challenges for accurate human detection and tracking. This paper presents a complete edge-computing solution that not only processes coarse thermal data to extract a reliable people count, but also performs the computation entirely on an ESP32 microcontroller, eliminating privacy risks associated with data transmission, while maintaining real-time performance. Our novel lightweight algorithmic pipeline employs linear interpolation, adaptive thresholding, blob feature extraction, centroid and central-points based tracking and, in addition, a persistence-based algorithm differentiates humans from static heat sources by analyzing micro-movement patterns over time, critically reducing false positives from subjects like lighting, heating, ventilation, and air conditioning (HVAC) systems in the field of view. The system's effective 3-D detection volume and counting capacity are first experimentally mapped to establish precise operational limits. The system is subsequently tested and validated across diverse real-world scenarios including a classroom, an outdoor courtyard area, an indoor confined space, and a library study pod. Our comprehensive evaluation measures count and occupancy accuracies separately over continuous 10-minute periods, achieving a longest consecutive count error of only 28 seconds in the pod. This temporal stability enables near-perfect reliability: any connected decision-making system with a 30-second window achieves correct decisions almost all the time, unaffected by brief count variations. By implementing all processing locally on an ESP32 (520KB SRAM), this work demonstrates that deterministic algorithms can provide a practical, scalable alternative to machine learning approaches for privacy-critical sensing applications, enabling easy deployment in real-world environments without reliance on cloud infrastructure or exposure to privacy risks.

INDEX TERMS People counting, thermal array sensor, privacy-preserving sensing, edge computing, AMG8833, human detection, occupancy monitoring, ESP32, grid-EYE sensor, embedded systems.

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I. INTRODUCTION

The pursuit of truly intelligent environments, from energy-efficient smart buildings to secure public spaces, has driven the development of systems that can intelligently manage

resources. A key aspect of such systems is their ability to be seamlessly responsive to occupants, and dynamically adapt to human presence and movement [1], [2], [3]. While conventional vision systems using RGB or depth cameras provide rich spatial data, they present several drawbacks: unreliable performance in variable lighting conditions, high hardware costs, and, most critically, substantial privacy infringements due to the capture of identifiable personal details. This inherent tension between functionality and privacy has catalyzed the search for alternative privacy-preserving sensing methods.

Low-resolution infrared (IR) thermal sensor arrays, such as the 8×8 pixel Panasonic Grid-EYE AMG8833 [4], have emerged as powerful alternatives. These sensors offer a robust, cost-effective, and inherently privacy-preserving solution by design. By capturing only coarse thermal signatures, they enable interpretation of key spatial occupancy data without recording any personally identifiable features. However, the very feature that guarantees privacy — the coarse low-resolution data — introduces a unique set of challenges, such as distinguishing faint human thermal patterns from ambient temperature fluctuations and robustly rejecting false positives caused by static heat sources like electronic equipment or radiators, to achieve reliably accurate performance.

Recent research has increasingly applied machine learning (ML) and deep learning (DL) approaches to identify patterns in these heat signatures and classify activities from such low-dimensional thermal data [5], [6], [7]. While powerful, these methods can introduce considerable overhead in terms of data acquisition for training, computational demands for model training and inference, and the risk of poor generalization to unseen data. For resource-constrained edge devices, where real-time performance, low power consumption, low costs, and predictability are paramount, such methods may prove impractical.

This paper diverges from this trend and proposes a deterministic algorithmic pipeline that addresses the constraints of a practical application. Our results demonstrate that a lightweight algorithm can achieve reliable performance without the complexities and resource burden of ML frameworks, while running in real time on embedded edge hardware. Our algorithm additionally integrates a novel persistence-based logic that robustly differentiates mobile human subjects from inanimate hot objects, a mechanism critical for mitigating false positives, a common failure point in thermal sensing systems. By performing all computations locally, our system further fortifies data privacy by eliminating the need for any data transmission, thus enabling a truly autonomous, standalone monitoring unit. The primary contributions of this work are therefore:

- A novel math-based algorithmic pipeline, encompassing interpolation, adaptive thresholding, blob analysis, and centroid-based tracking, that converts the coarse thermal input data into meaningful estimates of human presence and activity.

- The Integration of a persistence-based hot object differentiation algorithm that actively distinguishes stationary heat sources from human thermal signatures, significantly improving detection reliability.
- The successful implementation and validation of the complete processing pipeline directly on a resource-constrained ESP32 microcontroller, demonstrating a pathway to fully localized, privacy-preserving occupancy monitoring.
- A comprehensive experimental evaluation of the system across diverse indoor and outdoor scenarios, assessing its accuracy in people counting, robustness to environmental interference, and real-time operational viability.

The remainder of this paper is organized as follows: Section II reviews relevant literature on low-resolution thermal sensors and thermal-based human detection. Section III details the system architecture and the core algorithms. Section IV presents the experimental setup's results. Section V discusses the algorithm parameters and their optimization ranges. Finally, Section VI discusses the conclusions, and Section VII outlines limitations and directions for future work.

II. RELATED WORK

Low-resolution thermal sensor arrays have attracted significant interest in recent years for a multitude of applications, primarily due to their inherent privacy-preserving characteristics, operational resilience to varying light conditions, and cost-effectiveness. While their primary application revolves around human presence detection, human activity recognition (HAR), and tracking, the versatility of these sensors is demonstrated by their application in diverse fields. For instance, Rivai et al. explored their use in assessing egg freshness based on thermographic analysis [10], Fang et al. investigated guided super-resolution with IR for crack segmentation in masonry [11], and several studies have focused on static or dynamic hand gesture recognition [12], [13], [15]. For smart home applications, Chuah and Teoh designed a system for energy conservation that automatically controls appliances based on which section of a room a person occupies [14]. Furthermore, the unique thermal signatures captured by these sensors have been applied to specialized tasks such as fall detection in home elevators [16], [17] and the identification of anomalies in human gait (a person's manner of walking) [18]. This breadth of applications illustrates the versatility of low-resolution thermal imaging.

For human occupancy monitoring, researchers have employed a range of IR array sensors and associated processing platforms. The 8×8 pixel Panasonic AMG8833 is a frequent choice due to its low cost and wide availability [9], [19], [21], [22], [23]. Higher-resolution sensors, such as the 32×24 Melexis MLX90640 [13], [15], [16], [24], [25] and 32×32 Heimann sensors [12], [28], [33], are also utilized, with their increased spatial detail often leveraged for improved performance in specific tasks.

The choice of processing unit commonly includes PCs or single-board computers like the Raspberry Pi for deployment [9], [15], while research by Cevik et al. [33] explored STM32 microcontrollers, and our own work targets the even more resource-constrained ESP32.

Numerous studies have investigated the application of ML and DL techniques to interpret data from these thermal sensors. For the 8×8 AMG8833, Yin et al. demonstrated the use of Long Short-Term Memory (LSTM) networks for HAR [19] and later a Semi-Supervised Cross-Domain Neural Network (SCDNN) for environmental adaptability [23]. Mekruksavanich et al. developed LowResIR-Net, a lightweight residual network, for activities of daily living (ADL) recognition using AMG8833 data from the Coventry-2018 dataset [20], [21]. For the MLX90640, Muthukumar et al. designed a hybrid CNN-LSTM model with dual sensors for activity detection [24], and Yu et al. created ISFD-Net for fall detection in elevators [16]. Naser et al. adopted an ML approach with an artificial neural network (ANN) applied to features extracted from 32×24 MLX90640 sensor data to estimate human distance and height [25]. Rezaei et al. used a stereo setup with two MLX90640 sensors, and applied various ML models to demonstrate that fusing sensor data improves performance for activity and fall detection [26]. Geng et al., using a higher-resolution 80×62 thermal camera (Bobcat EVA), employed DBScan (Density-Based Spatial Clustering of Applications with Noise) for algorithmic head detection, subsequently using LightGBM (an ML model) on these extracted features for 2D position estimation [27].

Object detection frameworks have also been investigated; Klir et al. utilized raw 32×32 Heimann sensor images, augmented with synthetically generated data, to train YOLO/DETR models for localizing multiple persons and distinguishing them from thermal disturbances [28]. Laufer et al. provided a comparative analysis of various YOLO models for human detection with 32×32 thermal data, highlighting performance trade-offs [29], [30]. Some works also explore image enhancement techniques such as super-resolution, like Real-ESRGAN by Correa et al. [31] or diffusion models - DiffBIR by Chen et al. [12] to improve input data quality prior to DL-based processing, although these can introduce computational overhead. The processing for these ML/DL approaches typically occurs on PCs or powerful single-board computers, and demonstrates the power of data-driven models in extracting complex patterns. However, they often require significant datasets and computational resources for training [8], and their direct applicability to extremely resource-constrained devices like an ESP32 without substantial optimization or specialized architectures remains a challenge. Our own preliminary investigations using a fine-tuned YOLOv11 model, trained on 200 annotated images of blobs, failed to work on raw 8×8 AMG8833 data, posing issues with tracking stability due to bounding box jitter (as shown in Figure 1), even for stationary subjects [34], [35]. This steered our research towards a more deterministic approach.

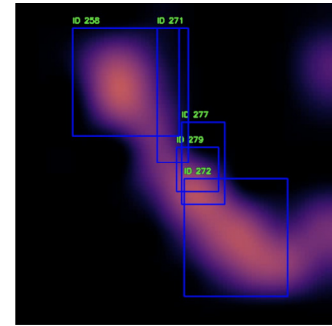


FIGURE 1. Bounding-box jitter in Yolo-based approach.

Complementing these methods, several studies have pursued direct algorithmic, or more deterministic approaches for processing low-resolution thermal data, often with a strong emphasis on computational efficiency and suitability for on-device implementation. Shubha and Shastrimath presented a real-time occupancy detection system using an 8×8 AMG8833, where the entire pipeline, including bicubic interpolation, adaptive thresholding, and Gaussian filtering, was executed on a Raspberry Pi [9]. Lin and Zhao also detailed a comprehensive math-based algorithm for human occupancy monitoring and positioning, leveraging the AMG8833 [32]. Newaz and Hanada proposed a unique approach for fall detection on a standalone device by analyzing statistical distributions of summed thermal signatures from an AMG8833 [22]. Similarly, Cevik et al. developed a human detection and tracking system for 32×32 IR images using a simple centroid displacement calculation, implemented using a STM32F401 board, explicitly choosing this over more complex filters to maintain efficiency [33]. These algorithmic strategies often prioritize edge deployment and speed, making them viable candidates for deeply embedded systems.

Our research builds upon this existing body of work. While inspired by the privacy-preserving nature of thermal sensors highlighted across the literature, and comparing the performances of both ML/DL and algorithmic techniques, this paper focuses on developing a robust and computationally lightweight deterministic algorithm. Further, we demonstrate the successful porting of the entire algorithm onto an ESP32 microcontroller, offering a practical solution unit, addressing the resource limitations of many real-world applications.

III. SYSTEM DESIGN AND ALGORITHM

This Section details the systematic design and progressive implementation of the proposed system for privacy-preserving human detection, tracking, counting, and static hot object differentiation. The system evolved from an initial server-facilitated architecture, used for rapid prototyping and algorithm development, to finally a fully localized and computationally efficient embedded solution unit. The sensor specifications, data acquisition protocols, and the algorithmic pipeline are detailed herein. The system architecture, as illustrated in Figure 2, evolved through a two-stage

development process aimed at achieving both robust algorithm performance and efficient embedded deployment.

A. SYSTEM ARCHITECTURE AND DEVELOPMENT WORKFLOW

The initial development phase (Figure 2, top) focused on algorithm design and validation using a server-centric approach. In this setup, a Grid-EYE thermal sensor, interfaced with an ESP32 microcontroller, acquired raw 8×8 thermal data, along with temperature measurement from a DHT Sensor. The ESP32 acted as an HTTP client, transmitting this data as JSON packets over a WiFi network to a server computer. At the server, this data was then processed using a Python script that implemented the algorithm (detailed in the later part of this Section) to obtain the occupancy status and count, and visualization of the blobs. This configuration allowed for rapid prototyping, algorithm testing, and detailed result validation using the server's computational resources.

Following successful algorithm development and validation on the server, the second phase (Figure 2, bottom) involved tailoring and optimizing the Python script for local execution on the ESP32 microcontroller. This transition required refactoring the code to remove non-essential functions (such as extensive visualization routines), minimizing computational overhead, and scaling the algorithms to operate within the ESP32's resource constraints. In this embedded configuration, the ESP32 directly performs the human detection and counting algorithm, with the output indicated via an onboard LED, thus creating a standalone, privacy-protected system. The initial server-client communication developed was preserved and used to test/log the standalone system when needed.

B. SENSOR SELECTION AND DATA ACQUISITION

Understanding the sensor's specifications is crucial for extracting meaningful information from its raw data and ensuring reliable long-term operation.

1) INFRARED THERMAL ARRAY SENSOR: AMG8833

For this research, the Panasonic Grid-EYE AMG8833 infrared thermal array sensor was selected. This sensor is well-suited for low-cost, privacy-conscious human detection applications. The AMG8833 is a thermopile array sensor that detects infrared radiation emitted by objects within its field of view (FoV), translating this into a two-dimensional thermal map of 8×8 pixels without capturing any visual features that could compromise individual identity [4]. Each of the 64 pixels measures the temperature of a corresponding section of the scene. The sensor typically operates with a temperature detection range suitable for human body temperatures and common indoor environments - 0°C to 80°C . A Key specification is the 60° viewing angle, which is sufficient for tracking human movement. The sensor outputs digital temperature values for each pixel via an I²C communication interface. Its compact package and low power

consumption facilitate integration with microcontrollers such as the ESP32.

While more advanced higher-resolution thermal sensors offer improved spatial detail, they present three critical challenges for privacy-preserving edge deployment. First, increased pixel resolution captures finer thermal features, compromising privacy by revealing greater anatomical detail. Second, computational demands scale proportionally with pixel count, surpassing the memory and processing limits of resource-constrained platforms like the ESP32. Third, advanced sensors incur significantly higher costs and power consumption (Table 1), undermining scalability for widespread deployment in cost-sensitive applications. The AMG8833 therefore represents the optimal balance between privacy preservation, computational feasibility, and deployment cost for edge-based deployment.

TABLE 1. Comparison of different thermal sensor specifications.

Sensor	Resolution	Total Pixels	Current (mA) at 3.3V
AMG8833 [4]	8×8	64	4.5
HTPA16x16d [36]	16×16	256	3.9
MLX90640 [37]	32×24	768	20
HTPA32x32d [38]	32×32	1024	6.2

2) WIRELESS DATA TRANSMISSION

During the initial algorithm development phase, a wireless mechanism was established to transfer raw thermal data acquired by the ESP32 to a host computer(Laptop) for processing and visualization. After acquiring the 8×8 thermal data array from the AMG8833 sensor via I²C, the ESP32 HTTP client periodically initiated POST requests to a Flask-based HTTP server running as a Python script on the Laptop. With each request, the ESP32 transmitted the latest captured thermal data array, along with a co-located DHT11 temperature reading, formatted as a JSON packet. This DHT11 reading serves as a reference point for the detection algorithm and is discussed further in Section III-C2. The data transmission frequency was optimized through several iterations to balance the demand for near real-time data updates with network stability and the ESP32's capacity as an HTTP client.

C. MATH-BASED DETECTION AND TRACKING ALGORITHM

The core of this human detection and tracking system employs a deterministic algorithmic pipeline. This pipeline processes raw 8×8 thermal data from the Grid-EYE sensor, through five main stages: data up-scaling, dynamic thresholding, blob feature extraction, multi-object tracking, and finally, normalization and visualization for output.

1) DATA UPSCALING: TWO-STAGE LINEAR INTERPOLATION TO 71×71

The native 8×8 pixel resolution of the AMG8833 sensor provides limited spatial detail, which can hinder effective

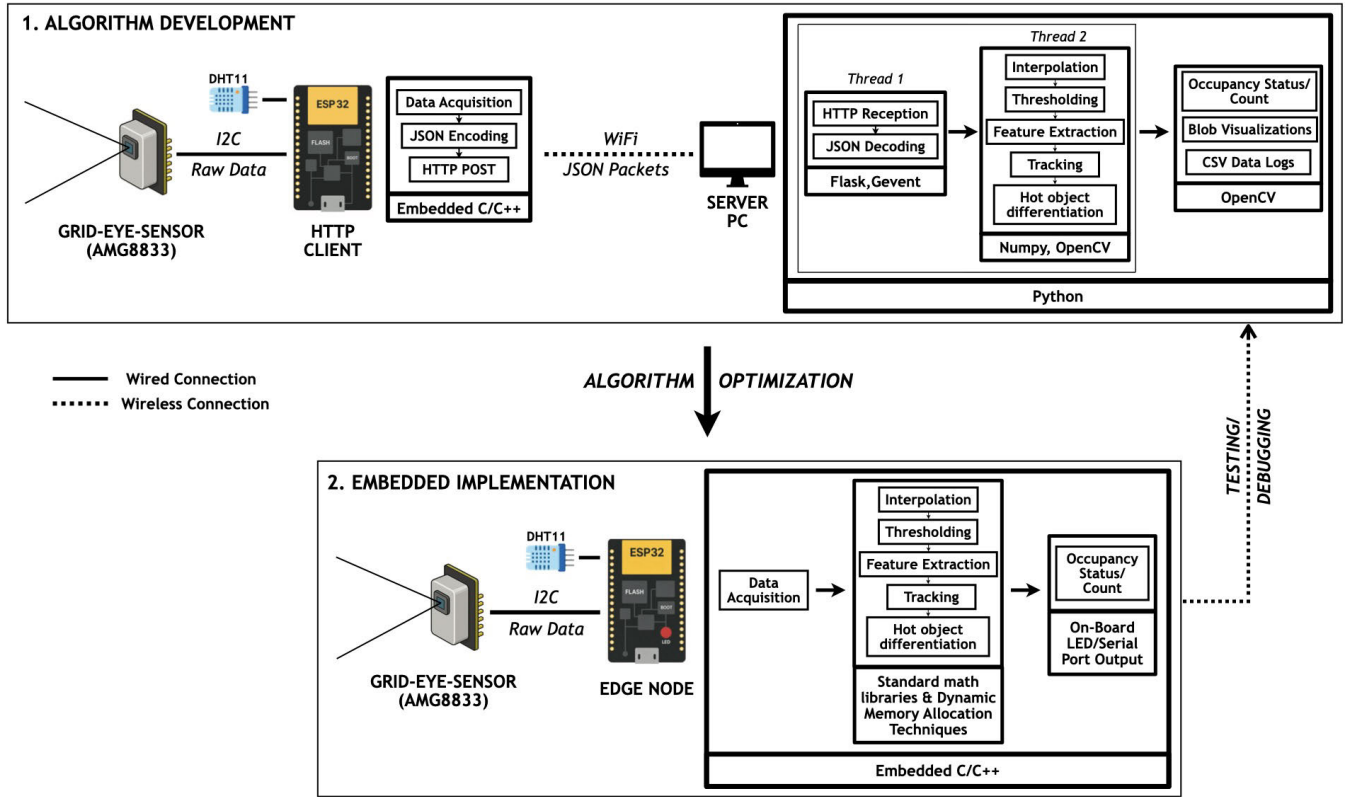


FIGURE 2. Two-stage system development. (Top) Server-dependent configuration for algorithm refinement. (Bottom) Final embedded system with local processing on ESP32 for enhanced privacy. (Icons generated using OpenAI's GPT-4o).

feature extraction. To address this, an initial up-scaling stage is implemented using two-stage linear interpolation. This process increases the resolution by inserting 9 new pixels between each pair of original adjacent pixels, applied both in the horizontal and the vertical dimension. This transforms the 8×8 input into a 71×71 up-scaled thermal map, I_{out} , as each 8-pixel line expands to $8 + (7 \times 9) = 71$ pixels.

The value of an interpolated pixel, $I_{out}(p_{new})$, situated on the line segment defined by two adjacent pixels with values $I_{in}(p_1)$ and $I_{in}(p_2)$, is determined by Equation 1.

$$I_{out}(p_{new}) = (1 - w)I_{in}(p_1) + wI_{in}(p_2),$$

$$\text{where, } w = \frac{k}{10}, \quad k \in \{0, \dots, 10\} \quad (1)$$

Here, w represents the normalized fractional distance of p_{new} from p_1 along the segment p_1p_2 . A value of $w = 0$ returns $I_{in}(p_1)$, while $w = 1$ returns $I_{in}(p_2)$, with intermediate w yielding proportionally interpolated values. The resulting 71×71 up-scaled image I_{out} provides a smoother thermal representation with enhanced effective detail for subsequent processing stages.

While more complex interpolation techniques exist, linear interpolation was selected for this work as it offers a sufficient enhancement for blob analysis from the coarse 8×8 input, while critically minimizing computational overhead, a key consideration for the target ESP32 deployment.

2) DYNAMIC THRESHOLDING AND MASK GENERATION

Effective segmentation of human thermal signatures from the background is a challenging task due to variations in ambient temperature across different environments over time. A fixed temperature threshold for binarization would necessitate frequent recalibration and would likely fail as conditions change. To address this and ensure robust performance, our system employs an adaptive dynamic thresholding mechanism.

This mechanism utilizes a co-located DHT11 temperature sensor next to the AMG8833 to provide a continuous, real-time measurement of the ambient room temperature, denoted as T_{DHT} . This ambient reading serves as a dynamic baseline for differentiating foreground (human) heat signatures. Through empirical testing with various offsets above and below this ambient baseline, an optimal thresholding strategy was determined (Equation 2). The threshold, $T_{threshold}$, is set slightly above the ambient temperature:

$$T_{threshold} = T_{DHT} + 0.1^\circ \text{C} \quad (2)$$

This small positive offset was found to provide a good balance, effectively segmenting human thermal profiles while minimizing noise from minor background thermal fluctuations, across the diverse conditions tested.

The 71×71 interpolated temperature map, $I_{out}(x, y)$, is binarized against this dynamic $T_{threshold}$ to generate a

binary mask $M(x, y)$, as defined in Equation 3:

$$M(x, y) = \begin{cases} 1, & \text{if } I_{out}(x, y) \geq T_{threshold} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

This binarization process classifies each pixel in the upscaled image as either foreground (1, considered part of a warm object, typically human) or background (0). The resulting binary mask $M(x, y)$ serves as the input for the subsequent blob feature extraction stage. This adaptive approach is crucial for maintaining consistent segmentation and robust performance regardless of the sensor's deployment location or temporal variations in room temperature.

3) BLOB FEATURE EXTRACTION

Following the generation of the binary mask $M(x, y)$ from dynamic thresholding, connected-component analysis (CCA), using OpenCV is applied. CCA identifies and labels distinct contiguous regions of foreground pixels, each forming a "blob" that represents a potential human or warm object. For each detected blob k , a set of descriptive features is extracted to characterize its properties for subsequent tracking and analysis:

- 1) **Area (A_k):** Denoted by A_k , is the total count of pixels constituting blob k .
- 2) **Centroid ($C_k = (\bar{x}_k, \bar{y}_k)$):** The geometric center, or center of mass, of blob k is computed as the average of the x-coordinates and y-coordinates of all pixels (x, y) belonging to the blob, as defined in Equation 4:

$$(\bar{x}_k, \bar{y}_k) = \left(\frac{1}{A_k} \sum_{(x,y) \in \text{blob}_k} x, \frac{1}{A_k} \sum_{(x,y) \in \text{blob}_k} y \right) \quad (4)$$

This centroid C_k provides a single-point representation of the blob's central location and serves as the primary spatial feature for the tracking algorithm described in Section III-C4.

- 3) **Central Points (CP_k):** To achieve a more robust characterization of a blob's core structure, especially useful for handling irregularly shaped or merged thermal signatures where a simple centroid might be misleading, a set of "central points", denoted by CP_k is identified. This process begins by computing a Manhattan distance transform, $D(x, y)$, over the binary mask of blob k . For each foreground pixel (x, y) within the blob, $D(x, y)$ represents its L_1 distance (sum of absolute differences in coordinates) to the nearest background pixel, as defined in Equation 5:

$$D(x, y) = \min_{(u,v) \in \text{background}} (|x - u| + |y - v|) \quad (5)$$

Pixels located deeper within the blob, further from its boundary, will yield higher $D(x, y)$ values, as per Equation 5.

To identify the most central regions within each blob, the distance transform values $D(x, y)$ are convolved

with a 3×3 Laplacian-like kernel K to detect local maxima, as defined in Equation 6:

$$K = \begin{bmatrix} -1 & -2 & -1 \\ -2 & 12 & -2 \\ -1 & -2 & -1 \end{bmatrix} \quad (6)$$

This kernel design emphasizes local peaks by assigning a strong positive weight (12) to the center pixel while applying negative weights to surrounding neighbors, effectively implementing a second-derivative operator that responds maximally to regions of highest centrality. The convolution operation $C(x, y) = D(x, y) * K$ produces high positive values at pixels representing the geometric centers of blob regions. Points satisfying dual criteria—distance transform values $D(x, y) > \alpha_{dt}$ and convolution responses $C(x, y) \geq \beta_{conv}$ —are designated as central points CP_k , ensuring both sufficient distance from blob boundaries and local optimality within the blob structure. These threshold values were determined through validation testing on representative thermal data to balance detection sensitivity while minimizing false positives from noise and edge artifacts. They were found to be 3 for α_{dt} and 6 for β_{conv} .

This set, CP_k , effectively offers a skeleton-like representation of the blob's warmest or most central regions and is utilized as a secondary feature in the tracking association logic to refine matches, particularly when centroid-based matching is ambiguous.

The extracted features for each blob in the current frame—namely its area A_k , centroid C_k , and set of central points CP_k —form the comprehensive input to the multi-object tracking stage.

4) MULTI-OBJECT TRACKING AND ASSOCIATION

To ensure detected thermal blobs are correctly associated with the same individuals from one frame to the next, and to provide an accurate occupant count, a multi-stage association logic is employed. This process evaluates each blob identified in the current frame, characterized by its centroid C_{new} and set of central points CP_{new} (as extracted in the previous Section III-C3), against existing active tracks.

- 1) **Primary Matching (Centroid-Based):** The initial association attempt relies on centroid proximity, a computationally inexpensive measure. For a new blob and an existing track j with last known centroid $C_{oldj} = (x_{oldj}, y_{oldj})$, their Manhattan distance is calculated as

$$d_j = |x_{new} - x_{oldj}| + |y_{new} - y_{oldj}| \quad (7)$$

A match is established with the track j yielding the minimum d_j , provided this distance is less than a pre-defined τ_c . This threshold defines the maximum spatial displacement allowed for a blob to be considered a continuation of an existing track based on its overall central position.

- 2) **Secondary Matching (Central Point Jaccard Index - Fallback):** If a new blob cannot be matched to an existing track via centroid proximity, or if an existing track remains unassociated after the primary matching stage, a secondary criterion based on structural similarity is applied. This fallback mechanism utilizes the Jaccard Index, J_j , to compare the set of central points of the unmatched new blob (CP_{new}) with the set of last known central points of an unmatched existing track j (CP_{old_j}), as given by:

$$J_j = \frac{|\text{Pairs}(CP_{new}, CP_{old_j})|}{|CP_{new}| + |CP_{old_j}| - |\text{Pairs}(CP_{new}, CP_{old_j})|} \quad (8)$$

Here, $\text{Pairs}(CP_{new}, CP_{old_j})$ denotes the set of point correspondences obtained by greedily matching each point in CP_{new} to the nearest unmatched point in CP_{old_j} within a Euclidean distance less than τ_d . A match is declared if J_j exceeds a specified τ_J . A high Jaccard Index indicates significant overlap in the core thermal structures of the new blob and the old track, supporting association even if centroids have shifted beyond the primary matching threshold. This secondary check is particularly beneficial for maintaining track continuity when thermal blobs temporarily merge, split, or undergo non-rigid deformations, where centroid positions might be unreliable but the underlying core thermal signature remains partially consistent.

- 3) **Track Management:** Based on the association outcomes, If a blob is successfully matched to an existing track (by either centroid or central point criteria), the track's features (centroid, size, set of central points, and path history) are updated with the current blob's information.

If a blob cannot be associated with any existing track, it initiates a new track.

Tracks that fail to be associated with any new detection for a specified number of consecutive frames (e.g., 3 frames, monitored by a virtual age counter variable) are deemed stale and are pruned from the list of active tracks. This mechanism prevents the accumulation of obsolete tracks and allows the system to adapt to objects leaving the sensor's field of view.

The total number of currently active tracks, after excluding any identified as persistent static hot objects (as detailed later in Section III-D), provides the real-time people count.

5) NORMALIZATION, VISUALIZATION, AND OUTPUT

To facilitate real-time monitoring and debugging during the server-based development phase, a comprehensive visualization pipeline was implemented using OpenCV. A critical step in effectively visualizing the low-contrast 71×71 upscaled thermal image, I_{out} , was its normalization and color mapping.

For enhanced visual contrast prior to color mapping, the pixel values of the 71×71 upscaled thermal image, I_{out} , were normalized to an 8-bit range (0–255). This normalization

established a dynamic range for visualization where the minimum intensity, T_{min_viz} , was set to a value slightly below the ambient temperature baseline, and the maximum intensity, T_{max_viz} , was taken from the current frame's maximum pixel value. Specifically, T_{min_viz} and T_{max_viz} were defined as:

$$T_{min_viz} = T_{DHT} - \Delta T_{offset}, \quad (9)$$

$$T_{max_viz} = \max_{x,y} I_{out}(x, y). \quad (10)$$

Here, T_{DHT} is the ambient temperature from the DHT11 sensor, and ΔT_{offset} is an empirically determined offset (e.g., 8°C).

Each pixel is then mapped to an 8-bit value using a linear transform with clamping:

$$s(x, y) = 255 \times \frac{I_{out}(x, y) - T_{min_viz}}{T_{max_viz} - T_{min_viz}}$$

$$I_{norm}(x, y) = \text{clip}(s(x, y), 0, 255) \quad (11)$$

where $\text{clip}(v, 0, 255) = \min(\max(v, 0), 255)$.

In the degenerate cases where the visualization bounds are invalid, the system prints a calibration error and displays a black visualization screen: specifically, when $T_{min_viz} \geq T_{max_viz}$, the frame is rendered as black and a calibration-check message is issued (e.g., "Calibration error: Check DHT sensor / AMG8833"). The normalized 8-bit image is subsequently rendered using a custom thermal colormap that transitions from white (coolest) through cyan hues to increasingly intense pinks (warmest), accentuating subtle thermal variations relevant to human presence.

The system generated 6 synchronized OpenCV windows to display the progression of the algorithm, as shown in Figure 3:

- **Stage 1: Raw Thermal Image:** Displays the unprocessed sensor data, after normalization and application of the custom colormap.
- **Stage 2: Interpolated Thermal Image:** Displays the raw 71×71 upscaled thermal image.
- **Stage 3: Segmented and Background Subtracted Threshold Mask:** Showcases the binary mask resulting from dynamic thresholding.
- **Stage 4: Blob Feature Visualization:** Overlaid onto the Stage 2 view (or a colormapped version), this stage displays the detected blob boundaries, their calculated centroids (green circles), and the identified central points (yellow dots) for each distinct thermal signature.
- **Stage 5: Object Tracking Visualization:** Building upon Stage 4, this view adds the historical paths of the tracked centroids (blue lines), illustrating the motion trajectories of detected objects over time.
- **Stage 6: Final Output with Hot Object Suppression:** Presents the thermal image after the differentiation and visual suppression (masking) of identified static hot objects (as detailed in Section III-D). If no hot objects are identified, then no blocked masks are present, as in the case shown in the Figure above.

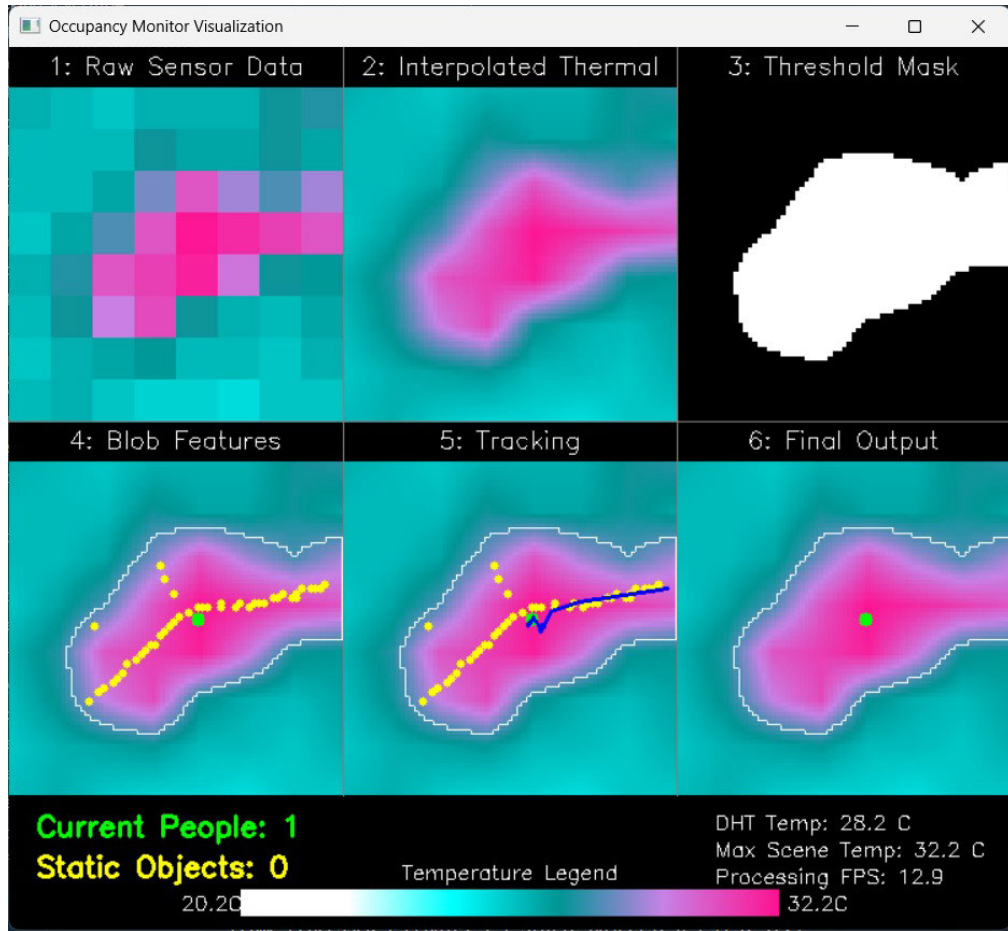


FIGURE 3. Visualization of the step-by-step algorithmic pipeline. The output window displays the data progression for a single occupant: (1) Raw 8×8 sensor data; (2) 71×71 interpolated thermal image; (3) Binary threshold mask; (4) Extracted blob features, including boundary (white), central points (yellow) and centroid (green); (5) The addition of the tracked path (blue); and (6) The final processed output with hot object suppression (not active in this frame). Status indicators at the bottom display the final people count and other real-time metrics.

The primary numerical outputs of the system, displayed alongside these visualizations and logged, were the real-time count of currently detected human occupants and the count of identified static hot objects.

D. HOT OBJECT DIFFERENTIATION USING PERSISTENCE ANALYSIS

A significant challenge in thermal-sensor based human occupancy monitoring is the potential misidentification of static heat sources like electronic devices such as charging adapters and batteries, lighting devices like bulbs, sun-heated surfaces, kitchen appliances, or AC vents and radiators as human presence. To address this, the system incorporates a persistence-based differentiation mechanism. The core principle relies on monitoring the movement characteristics of each tracked thermal blob over a sequence of frames, leveraging the observation that human subjects, even when attempting to remain still, typically exhibit micro-movements, whereas inanimate hot objects remain truly stationary for extended periods.

This differentiation process is executed after the initial tracking of thermal blobs (as detailed in Section III-C4). For each active track, which maintains a temporal history of centroid coordinates (x_t, y_t) and area A_t , the logic operates as follows:

1) Monitoring movement history and drift calculation:

For every blob not already classified as persistent, we evaluate its centroid displacement by comparing the current position to its position N_{window} frames prior, where N_{window} defines the temporal window size for drift detection. The drift, δ , is defined as the Euclidean distance between the current centroid $(x_{\text{current}}, y_{\text{current}})$ and the centroid recorded N_{window} frames earlier, $(x_{\text{past}}, y_{\text{past}})$:

$$\delta = \sqrt{(x_{\text{current}} - x_{\text{past}})^2 + (y_{\text{current}} - y_{\text{past}})^2} \quad (12)$$

The scalar δ represents the net displacement over the observation window and is evaluated every frame as the sliding endpoint displacement metric for the persistence test.

- 2) **Static Object Classification:** The algorithm implements a consecutive-pass counter for each blob: when δ is below the low-movement threshold Δ_{static} , the per-blob counter c_k , for each blob k , is incremented; otherwise the counter is reset to zero. Formally:

$$c_i = \begin{cases} c_i + 1, & \delta < \Delta_{\text{static}} \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

A blob is classified as a “static hot object” when $c \geq N_{\text{persist}}$, where N_{persist} represents the minimum number of consecutive frames required for persistence confirmation. Upon classification, the current centroid is recorded as the locked position $\mathbf{p}_{\text{locked}}$ and a circular binary mask M_{locked} is generated centered at this location. The mask radius is determined from the object’s area with additional dilation parameter ρ_{dilate} to ensure complete spatial coverage.

- 3) **Resetting persistence:** For blobs already classified as persistent, every new current centroid is continuously compared to the stored locked position $\mathbf{p}_{\text{locked}}$. If the Euclidean distance between these points exceeds the reset threshold Δ_{reset} , the persistent classification is removed and the blob reverts to a regular tracked state, eligible for future re-evaluation. This ensures objects that were temporarily stationary but subsequently move are re-evaluated and correctly re-classified. Additionally, when a blob is no longer detected by the tracker, its persistent entry is automatically cleared from the registry.
- 4) **Background suppression and occupancy counting:** At each frame, a combined mask is created by merging all persistent object masks. This combined mask M_{combined} identifies all pixels that belong to any static heat source:

$$M_{\text{combined}}(i, j) = \bigvee_{k=1}^{N_p} M_k(i, j) \quad (14)$$

where M_k represents the mask of persistent object k and N_p is the total number of persistent objects. The thermal image is then modified by replacing pixels covered by static objects with a baseline temperature value:

$$I_{\text{out}}(i, j) = \begin{cases} T_{\text{baseline}}, & \text{if } M_{\text{combined}}(i, j) = 1 \\ I_{\text{interp}}(i, j), & \text{otherwise} \end{cases} \quad (15)$$

This makes static heat sources invisible in the display, showing only moving thermal signatures. For occupancy counting, persistent blobs are excluded from the final count, ensuring only human presence is measured.

This persistence-based analysis provides a deterministic method to differentiate between human thermal signatures, which exhibit natural movement patterns, and static heat sources. The algorithm’s performance is governed by four key parameters: the consecutive frame requirement N_{persist} , the drift detection window N_{window} , the static movement

threshold Δ_{static} , and the persistence reset threshold Δ_{reset} . Proper calibration of these parameters balances sensitivity to human micro-movements with effective suppression of static thermal interference, enhancing occupancy detection accuracy.

E. LOCALIZED ESP32 IMPLEMENTATION

While the server-based architecture (Figure 2-Top) was instrumental for rapid algorithm development and visualization, a primary objective of this research was to transition towards a fully localized, inherently privacy-preserving, and resource-efficient embedded system. This was accomplished by implementing the core detection, tracking, and hot object differentiation algorithms directly on the ESP32 microcontroller (Figure 2-Bottom).

The initial role of the ESP32 was confined to data acquisition from the AMG8833 sensor and subsequent wireless transmission of raw thermal data to the laptop server, where all computationally intensive tasks were handled. An ESP32-WROOM-32 module (dual-core Xtensa LX6, 80–240 MHz) with 520 KB on-chip SRAM and 8 KB + 8 KB RTC SRAM was used [39]. To achieve on-device processing of the raw sensor values, the validated Python-based algorithms were systematically translated and adapted into an embedded C/C++ script tailored for the ESP32. This porting process involved several critical considerations:

- 1) **Algorithmic Simplification and Optimization:** Functionalities primarily related to visualization were removed. The focus was strictly on implementing the core mathematical operations essential for detection, tracking, hot object differentiation, and outputting an accurate occupancy count.
- 2) **Core Algorithm Porting:** The fundamental stages of the detection pipeline were re-implemented in embedded C/C++ for the ESP32 microcontroller environment. This included:
 - Two-stage linear interpolation transforming 8×8 raw sensor data to the 71×71 upscaled grid.
 - Dynamic thresholding implementation using ambient temperature baseline from the co-located DHT11 sensor.
 - Connected-component analysis for thermal blob identification and labeling.
 - Feature extraction including centroid computation $C_k = (\bar{x}_k, \bar{y}_k)$ and central points CP_k identification using distance transform and Laplacian kernel convolution.
 - Multi-object tracking with primary centroid-based association and secondary central point Jaccard Index matching for robust data association and stale track pruning.
 - Persistence-based static object differentiation incorporating drift calculation, consecutive-pass counting logic, and background suppression.

The embedded implementation directly outputs the real-time human occupancy count, with persistent

static objects automatically excluded from the final tally.

3) **Dynamic Memory Management for Resource Constraints:**

A critical challenge in algorithm porting was managing the ESP32's limited on-chip RAM capacity (520 KB SRAM and 8 KB + 8 KB RTC SRAM), particularly for intermediate data structures including the 71×71 interpolated thermal arrays and binary segmentation masks. To prevent memory overflow and ensure stable operation, a frame-synchronized dynamic allocation strategy was implemented:

- Large computational buffers required for per-frame processing—including the upscaled thermal matrix, binary mask arrays, and auxiliary data structures for connected-component analysis (pixel visitation arrays and breadth-first search queues)—were dynamically allocated at the start of each processing cycle. Raw thermal frames (acquired from the AMG8833 sensor via I²C) were passed directly into this pipeline to minimise intermediate buffering.
- These transient memory buffers were explicitly deallocated at the conclusion of each processing cycle, before the next frame acquisition, preventing memory accumulation across successive frames.
- Persistent tracking structures maintaining object trajectory histories were allocated dynamically during runtime upon detection of new thermal signatures and deallocated when objects become stale (exceed 3 frames without updates), with a maximum capacity constraint of $N_{max} = 10$ concurrent tracked objects.

This temporal memory management approach, with allocation-deallocation cycles synchronized to frame processing intervals, enabled execution of the complete multi-stage processing pipeline within the microcontroller's on-chip memory constraints while maintaining computational stability.

- ### 4) **On-Board Output Indication:**
- For immediate, standalone feedback, an on-board LED interfaced with the ESP32 provides a binary occupancy indication. The LED state transitions to active when one or more persons are detected ($N_{people} > 0$) and remains inactive otherwise, offering real-time occupancy status without requiring external communication. Detailed metrics including exact person count and persistent object status are computed internally and available for serial output or optional wireless transmission.

To facilitate verification and debugging of the localized ESP32 implementation, the capability to transmit raw sensor readings (or processed results) to the server was retained as an optional feature within the script. This allowed for concurrent operation where the ESP32 performed its local computations and LED indication, while simultaneously sending data to the laptop. The original Python script could then receive

these live readings for visualization and direct comparison, validating the on-board algorithm's performance against the server-based reference.

The ESP32's processing capability (dual-core Xtensa LX6, up to 240 MHz) proved sufficient to produce near-real-time occupancy outputs—LED indication occurred effectively immediately upon a subject entering the FoV. The system's power consumption was modest, drawing approximately 100–300 mA (0.5–1.5 W at 5 V), well within typical USB supply limits.

This localized implementation significantly enhances data privacy by processing thermal data directly on the edge device. It culminates in a truly standalone occupancy monitoring unit, capable of independent operation for its core detection task while mitigating memory constraints through careful dynamic memory management. The complete, optimized processing logic executed by the microcontroller is summarized in Algorithm 1.

Algorithm 1 Localized Embedded Detection Algorithm

```

1: Initialize:  $I^2C$ , GPIO, Track List  $\mathcal{T} \leftarrow \emptyset$ 
2: loop
3:   Acquire:  $I_{in}, T_{DHT}$ 
4:   Calibrate:  $T_{thresh} \leftarrow T_{DHT} + 0.1^\circ C$   $\triangleright$  Eq. (2)
5:   Upscale (Dynamic Allocation):
6:      $I_{out} \leftarrow \text{Interpolate}(I_{in}) 8 \times 8 \rightarrow 71 \times 71$   $\triangleright$  Eq. (1)
7:   Segment (Dynamic Allocation):
8:      $M(x, y) \leftarrow 1 \text{ if } I_{out}(x, y) \geq T_{thresh} \text{ else } 0$   $\triangleright$  Eq. (3)
9:   Feature Extraction:
10:     $\mathcal{B} \leftarrow \text{FloodFill}(M)$   $\triangleright$  Computes  $\{A_k, C_k\}$ 
11:    Compute  $CP_k$  via Distance Transform  $\triangleright$  Eq. (5, 6)
12:   Track Association: Match  $b \in \mathcal{B}$  to  $t \in \mathcal{T}$  satisfying:
       $\min(\text{dist}(C_b, C_t)) < \tau_c \text{ OR } \text{Jaccard}(CP_b, CP_t) > \tau_J$ 
13:   Persistence & Classification:  $\forall t \in \mathcal{T}$ 
14:      $\delta \leftarrow \|C_t^{curr} - C_t^{prev}\|$   $\triangleright$  Drift, Eq. (12)
15:      $c_t \leftarrow (c_t + 1) \text{ if } \delta < \Delta_{static} \text{ else } 0$   $\triangleright$  Eq. (13)
16:     if  $(c_t \geq N_{persist})$   $Status_t \leftarrow STATIC$ 
17:     if  $(Status_t \text{ is } STATIC \text{ and } \delta > \Delta_{reset})$ 
18:        $Status_t \leftarrow DYNAMIC$ 
19:   Output:
20:      $N_{people} \leftarrow \text{Count}\{t \in \mathcal{T} \mid Status_t = DYNAMIC\}$ 
21:     LED  $\leftarrow (N_{people} > 0)$ ; Serial:  $N_{people}, N_{static}$ 
22:   Cleanup: Deallocate  $I_{out}, M$   $\triangleright$  Free Heap Memory
23: end loop

```

IV. EXPERIMENTAL SETUP AND RESULTS

This section details the performance assessment approach and presents the evaluation results of the proposed low-resolution thermal occupancy monitoring system. The core algorithm was running locally on an ESP32 microcontroller, with simultaneous data transmission to a laptop for verification, detailed visualization, and data logging.

A. HARDWARE CONFIGURATION AND SYSTEM ASSEMBLY

The system utilized a Panasonic Grid-EYE AMG8833 8×8 thermal sensor interfaced with an ESP32 development board via I²C. A DHT11 temperature and humidity sensor was also connected to the ESP32 to provide ambient temperature readings, which were used by the on-device algorithm for dynamic thresholding. A green LED, controlled by the ESP32, served as a binary indicator of detected human presence. The complete system was assembled on a standard prototyping breadboard and powered by a portable USB power bank, ensuring mobility and deployment flexibility. Figure 4 illustrates the physical hardware setup. For data verification during testing, the ESP32 also transmitted sensor readings via HTTP over a local WiFi network to a laptop running the Python-based visualization script.

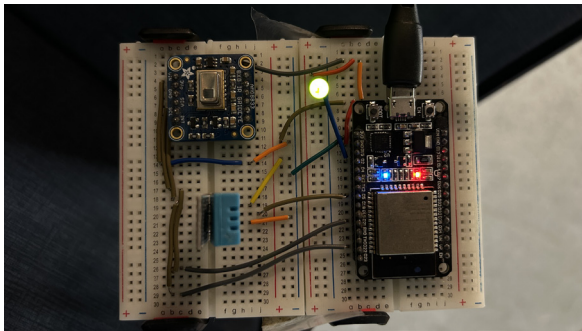


FIGURE 4. The breadboard assembly.

B. TEST SCENARIOS AND PERFORMANCE EVALUATION

The system's performance was evaluated through five experiments, designed to first establish a baseline characterization of its operational limits, and subsequently validate its accuracy and robustness in a variety of real-world scenarios. Ground truth occupancy counts were established through manual annotation of concurrently recorded RGB video footage, with timestamps, for comparison against the thermal system output. Informed consent was obtained from all participants involved in the experimental validation.

1) SCENARIO 1: INDOOR CLASSROOM: BASELINE CHARACTERIZATION

This experiment was conducted in a classroom (dimensions: $7\text{m} \times 9\text{m} \times 3.5\text{m}$), and measurements were acquired during afternoon hours (3:00 pm) under typical indoor environmental conditions, with all fans in the room turned off. The goal was to establish the system's baseline performance by mapping the effective detection volume (defining its range) and quantifying the minimum subject separation needed to prevent multiple-person blob merging.

- **Sensor Placement:** The ESP32 sensor unit was affixed to a wall at a height of 1.4 meters from the ground, facing an empty hallway between 2 rows of benches. This placement ensured the sensor's field of view was within

the average height of a human torso, and facilitated the measurement of detection boundaries at various distances, thereby mapping the effective 3-D Detection range.

- **Test Procedure:** To ensure clarity, a coordinate system is defined relative to the sensor: the Z-direction represents the forward line of sight, while the X and Y directions represent the horizontal (lateral) and vertical dimensions, respectively.

First, the maximum forward detection distance was determined. A single participant, initially positioned 6.0 meters away from the sensor, moved forward in 0.25-meter increments. At each marked position, the participant paused for 15 seconds to allow for thermal signature stabilization. The maximum distance, denoted as Z_{max} , was established as the furthest point from the sensor at which the system consistently reported a person count of 1. To ensure positional accuracy, care was taken to align the center of the participant's stance with each ground marking during the test.

Next, the horizontal and vertical extents of the detection volume were mapped on the plane at the established maximum distance Z_{max} . For the horizontal sweep, the participant moved laterally along the X-direction in 0.25-meter increments, pausing at each step for 15 seconds, first to the right and then to the left from the center, until detection was lost. To enhance the precision of these boundary measurements, this process was repeated with finer increments near the detected edge. This yielded the two boundary distances from the center, X_+ and X_- . For the vertical sweep, the participant, positioned at the center, sequentially assumed various postures (standing, crouching, sitting, lying down) and stood on stools of increasing height.

Finally, the system's capacity for multi-person counting was quantified. On the plane at $Z = Z_{max}$, two participants started at opposite ends X_+ and X_- and moved towards each other. The inter-person distance was decreased by 0.25-meters each time until their distinct thermal signatures merged into a single blob. This minimum distance required for successful discrimination was defined as the minimum spatial separation, S_{min} . The occupancy limit was then assessed by incrementally adding more participants into the scene, ensuring each maintained a separation of at least S_{min} from their neighbors, until the physical space within the detection volume could no longer accommodate another person while satisfying this constraint.

The first maximum forward detection experiment was further extended to characterize distance-dependent blob morphology, validate tracking robustness and effect of persistence-based masking during frontal approach along the Z-axis. As the participant advanced from $Z = Z_{max}$ m to $Z = 0.5$ m (directly in front of the sensor) in 0.25-meter decrements, thermal blob data was captured at each position, and analyzed.

- **Results:** The system successfully detected the participant up to a maximum forward distance of $Z_{max} = 3.0$ m. At this distance, the horizontal detection boundaries were measured as $X_+ = 1.55$ m and $X_- = 1.45$ m from the central position.

For the vertical sweep, the detection limits could not be directly measured, as the participant's thermal signature remained within the sensor's field of view across all tested postures, including the supine position and elevated stances. Given that the AMG8833 sensor exhibits identical horizontal and vertical viewing angles of 60° [4], the effective vertical range can be inferred from the measured horizontal extent. The total horizontal viewing angle, θ_{total} , was calculated as:

$$\theta_+ = \arctan\left(\frac{X_+}{Z_{max}}\right) = \arctan\left(\frac{1.55}{3.0}\right) \approx 27.3^\circ$$

$$\theta_- = \arctan\left(\frac{X_-}{Z_{max}}\right) = \arctan\left(\frac{1.45}{3.0}\right) \approx 25.8^\circ$$

$$\therefore \theta_{total} = \theta_+ + \theta_- \approx 53.1^\circ \quad (16)$$

This angle of approximately 53.1° is consistent with the sensor's specified 60° viewing angle ($\pm 30^\circ$ from the optical axis).

Figure 5 visualizes the effective 3-D detection volume. The system successfully identifies a thermal blob for any human subject within this frustum, even at the edges, provided the subject stands centered on the boundary line or at least 50% of their body is present inside the volume.

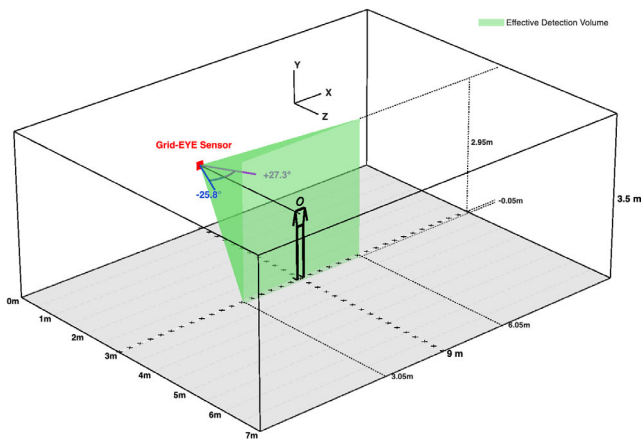


FIGURE 5. Visualization of the experimentally mapped effective detection volume. The green frustum represents the operational zone where a human subject generates a valid thermal blob, bounded by the measured horizontal/vertical viewing angle of 53.1° relative to the sensor.

The minimum spatial separation required for distinguishing two individuals was determined to be $S_{min} = 0.7$ m. With this spacing constraint, the system successfully discriminated up to four occupants positioned laterally across the detection plane at $Z = 3$ m, as shown in Figure 6. The maximum capacity

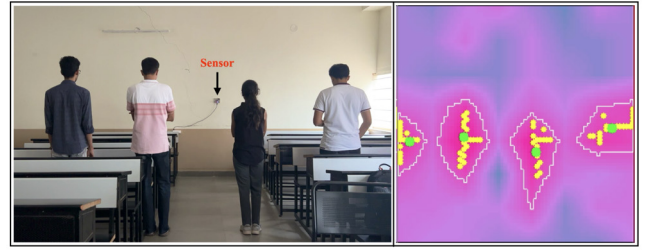


FIGURE 6. Evaluation of maximum detection capacity. (Left) Four participants aligned at the 3m detection plane, separated by the measured $S_{min} = 0.7$ m. (Right) Four distinct blobs correctly identified by the system.

was geometrically constrained: accommodating five individuals would require a total span of $5 \times 0.7 = 3.5$ m, exceeding the available detection width of $X_+ + X_- = 3.0$ m, which was then experimentally confirmed when the addition of a fifth individual resulted in thermal blobs merging permanently.

The frontal approach experiment validated the robustness of the system. Analysis of the blob visualizations at each position (representative samples shown in Figure 7) showed an average centroid shift of 20.77 pixels (Manhattan distance) between 2 consecutive visualizations. Since this value is less than the primary threshold ($\tau_c = 25$ pixels), primary matching would typically succeed, with the secondary, central-point-based matching acting as a robust fallback for larger shifts. Concurrently, the average Euclidean drift (δ) of the centroids over these same visualizations was 15.92 pixels. This is 36 times larger than the Δ_{static} (0.5 pixels) threshold, definitively preventing the misclassification of a stationary human. Figure 8 (a) illustrates the centroid trajectory.

The thermal blob's horizontal extent was observed to increase systematically with proximity. This was quantified using a coverage ratio $\rho(Z) = B_{pixels}(Z)/W_{pixels}$, where B_{pixels} is the maximum blob width and $W_{pixels} = 442$ pixels is the total field width, both measured in pixels from the system's graphical visualization output. Measured coverage ratios increased from $\rho(3.0 \text{ m}) = 0.231$ to $\rho(0.5 \text{ m}) = 0.995$, as shown in Figure 8(b). This pronounced inverse relationship between distance and blob coverage directly constrains multi-occupant capacity at closer ranges. Calculations show this critical distance is $Z = 1.5$ m, after which the coverage ratio

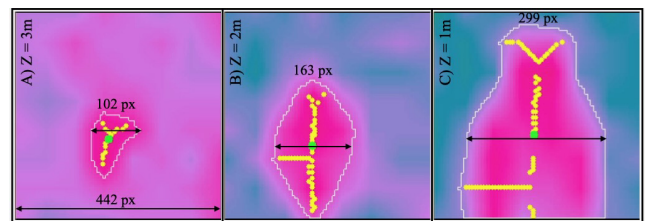


FIGURE 7. Evolution of thermal blob morphology during frontal approach. Blob visualizations at (A) $Z = 3$ m, (B) $Z = 2$ m and (C) $Z = 1$ m, illustrate the significant expansion of blob width (in pixels) relative to the fixed 442-pixel field width as the subject approaches the sensor.

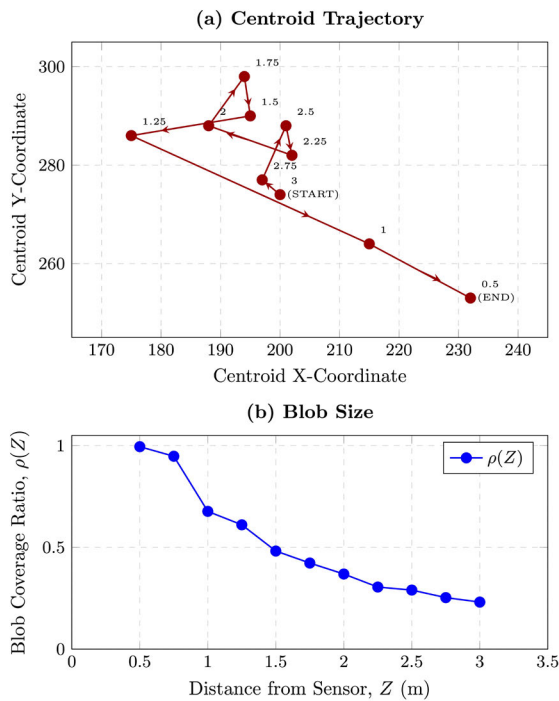


FIGURE 8. (a) The centroid's (x, y) pixel path as the participant moves from $Z = 3\text{m}$ (START) to $Z = 0.5\text{m}$ (END), with points labeled by distance Z in meters. (b) The blob size increasing as a function of distance Z .

$\rho(Z)$ exceeds 0.5. This implies a single blob occupies more than half the detection field, causing blobs to merge if a second individual enters. Consequently, while the exact people count is compromised at this proximity, the binary occupancy detection (count > 0) remains correct. The system maintains consistent tracking performance for any blob, regardless of the number of active subjects, provided these geometric limits are satisfied.

2) SCENARIO 2: INDOOR CLASSROOM WITH OVERHEAD SENSOR MOUNTING AND VARIABLE AIRFLOW

This experiment was conducted in a classroom (dimensions: $7\text{m} \times 9\text{m} \times 3.5\text{m}$) to assess multi-person detection

accuracy under varying airflow conditions. Measurements were acquired during afternoon hours (3:00 pm) under typical indoor environmental conditions.

- **Sensor Placement:** The ESP32 sensor unit was mounted on an overhead projector at a height of 2.5 meters from the ground.
- **Test Procedure:** The system was tested with one, two, and three participants sequentially entering the monitored area. Six trials were conducted for each participant count: three with two ceiling fans off and three with the fans operating at medium speed, as shown in figure 9.
- **Results:** The system demonstrated high accuracy in people counting. In scenarios with fans off, all three individuals were consistently identified and tracked correctly in all three trials. With fans operating, the system precisely identified and tracked all three individuals in 2 out of 3 trials. Instances of temporary blob merging were observed when participants were in very close proximity; however, the system typically maintained a people count ≥ 1 during such events and correctly re-segmented individuals as they separated. The overall people count accuracy in this classroom environment consistently exceeded 94%.

3) SCENARIO 3: OUTDOOR COURTYARD WITH STATIC HEAT SOURCE INTERFERENCE

This scenario was designed to evaluate the effectiveness of the hot object differentiation algorithm. Data collection was conducted during midday hours (12 noon) under normal ambient conditions.

- **Sensor Placement:** The sensor unit was affixed to an external building wall, 2 meters above ground, overlooking an open courtyard, with an exhaust vent in the field of view.
- **Test Procedure:** An active air-conditioning (AC) exhaust vent ($\sim 40^\circ\text{C}$ surface temperature) served as a persistent static heat source ~ 1 meter away from the sensor's position. A single participant entered the frame and remained as still as possible for a period exceeding the static object detection threshold. Five trials were

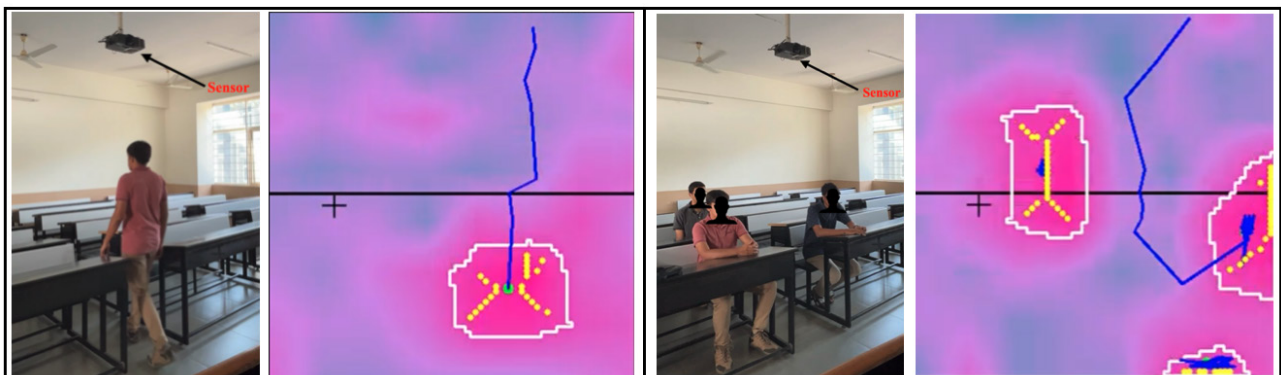


FIGURE 9. RGB camera views and corresponding thermal blob detection outputs for classroom scenario: single-person detection (left) and two-person detection (right) with tracked centroids, central points, and trajectories.

conducted. The hot object differentiation parameters were set as: $N_{persist} = 100$ frames (10 seconds), $\Delta_{static} = 0.5$ pixels, and $\Delta_{reset} = 2.0$ pixels. Figure 10 shows the test setup and the corresponding visualizations.

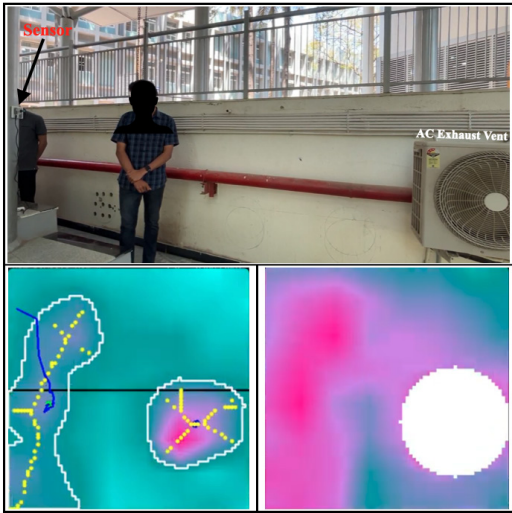


FIGURE 10. Outdoor scenario showing RGB view (top) and thermal detection with static object masking of AC exhaust vent (bottom).

- *Results:* The hot object differentiation mechanism successfully identified and masked the AC vent as a static object in 3 out of 5 trials after the 10-second persistence period, while maintaining accurate tracking of the stationary human participant. In one trial, the participant’s prolonged stillness resulted in temporary misclassification as a static object. In the remaining trial, thermal fluctuations from the AC vent exceeded the movement threshold Δ_{static} , likely due to external environmental factors such as winds or airflow variations typical in outdoor deployments, preventing static classification and resulting in continued detection as a dynamic object. Thus, the overall system performance yielded a False Positive Rate of 20% for human misclassification and a True Positive Rate of 60% for static object detection under these challenging environmental conditions.

4) SCENARIO 4: INDOOR ROOM WITH OBLIQUE SENSOR MOUNTING

This scenario aimed to determine the system’s effective detection range and performance with non-orthogonal sensor viewing angles in a smaller room (dimensions: 2.75m width $\times$ 5m length $\times$ 2.75m height).

- *Sensor Placement:* The sensor unit was mounted on a curtain rod on a wall at a height of 2.14 meters from the floor and tilted downwards at approximately 30 degrees (as shown in Figure 11).
- *Test Procedure:* One and two participants moved at varying distances and lateral offsets from the sensor.



FIGURE 11. Indoor room scenario showing oblique sensor mounting (left) and participant at varying distances for range testing (right).

21 distinct floor positions were evaluated within the floor’s area.

- *Results:* The system accurately detected and counted individuals up to an effective range of approximately 2.625 meters. Beyond this distance, the reduction in the temperature difference between the human subjects and the ambient room temperature made robust blob segmentation increasingly challenging, and the prediction failed. No blobs were identified, or visible in the visualization. For two-person detection across varying distances, systematic measurements were conducted using different configurations at each distance across three lateral positions, as detailed in Table 2. In the table, (X) indicates occupied positions and (.) indicates vacant positions, showing adjacent (X X .), (. X X) and separated (X . X) person arrangements.

TABLE 2. Two-person detection outcomes in confined room. Position configurations show occupied (X) and vacant (.) positions across three lateral locations at each distance from the sensor.

Vertical Distance (in metres)	Position Configuration (Horizontal Distance)			Actual People	Detected People
	Pos 1 (0.75m)	Pos 2 (1.5m)	Pos 3 (2.25m)		
0.75	X	X	.	2	2
	X	.	X	2	2
	.	X	X	2	1
1.5	X	X	.	2	2
	X	.	X	2	2
	.	X	X	2	2
2.25	X	X	.	2	2
	X	.	X	2	1
	.	X	X	2	1
2.625	X	X	.	2	1
	X	.	X	2	2
	.	X	X	2	1
3	X	X	.	2	0
	X	.	X	2	0
	.	X	X	2	0

The system demonstrated optimal performance at intermediate ranges of 1.5-2.25m, achieving near-perfect accuracy. At closer proximity (0.75m), adjacent positioning caused thermal blob merging, keeping the detected count at 1 person, though human presence remained correctly identified. After 2.25m, Detection accuracy progressively degraded with distance, declining to 50% at the 2.625m range where individual blobs became increasingly difficult to distinguish from background thermal variations. Beyond 2.625m, complete detection failure occurred (0% accuracy) as thermal signatures fell below the dynamic segmentation threshold, establishing the effective operational range limit. This visualization is shown in Figure 12.

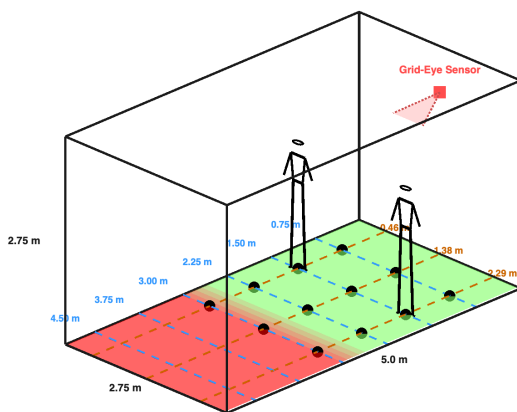


FIGURE 12. Visualization of indoor room test setup showing sensor placement, participant positions at various distances, and detection performance zones (green: reliable detection, red: detection failure, beyond 2.625m).

5) SCENARIO 5: CONFINED STUDY POD ENVIRONMENT

This scenario evaluated the system’s peak performance under favorable real-world conditions to measure the maximum achievable accuracy. Based on the previous test results and an understanding of the Grid-EYE sensor’s nature and limitations, the most suitable environment for thermal occupancy detection would require:

- Absence of natural wind or outdoor air disturbances,
- Controlled temperatures that do not vary significantly,
- A small, confined area that matches the sensor's limited detection range.

A study pod (a small, enclosed workspace designed for 1-2 individuals) in a library represents an ideal environment meeting all these requirements, providing the optimal conditions necessary for the Grid-EYE sensor to operate at its full potential, allowing determination of the algorithm's maximum achievable accuracy.

- *Environment:* Tests were conducted within a two-seater study pod located in an indoor library. Figure 13 shows the study pod environment.
- *Sensor Placement:* The sensor unit was positioned in an upper corner of the study pod via a suction-cup



FIGURE 13. The confined study pod environment (2.20m×1.05m×2.10m) utilized for Scenario 4 testing in a library.

mount attached to the glass partition. It was positioned at an angle such that it covers the entire interior space, as shown in Figure 14.



FIGURE 14. Sensor unit placement within the study pod.

- *Test Procedure:* Three distinct occupancy levels were evaluated, each for a continuous duration of 10 minutes, with the thermal data and corresponding blob visualizations being continuously logged and observed:
 - 1) Zero occupants (empty pod).
 - 2) One occupant seated inside the pod.
 - 3) Two occupants seated inside the pod.
- *Results and Analysis:* The logged thermal data and counts were validated against concurrent RGB video recordings. The key findings are summarized in Table 3. The table presents counting and detection performance across the three occupancy scenarios. The longest consecutive count error metric represents the maximum duration of continuous misclassification in exact person count in seconds. The longest consecutive occupancy error metric represents the maximum duration of continuous misclassification in binary occupancy status in seconds. The latter is particularly important for practical applications, as short periods can be easily managed by setting alert thresholds above the maximum observed duration. For instance, if the longest incorrect detection lasts 30 seconds, implementing a 60 second confirmation threshold before triggering occupancy alerts would

effectively eliminate false alarms, resulting in near-perfect system accuracy for real-world deployment.

- 1) For the **zero-occupant scenario**, the system demonstrated high reliability, achieving an accuracy of 92.02% in correctly identifying the pod as empty. The longest consecutive count error was only 1.05 seconds.
- 2) In the **one-occupant scenario**, the system correctly identified the presence of a single person with an accuracy of 77.12%. The longest consecutive count error period (Max time with a output count = 0) was 28.21 seconds.
- 3) The **two-occupant scenario** presented counting challenges, primarily due to the confined space leading to merged thermal blobs when individuals were in close proximity. The system correctly counted two persons 68.99% of the time, but achieved 100% occupancy detection as all predictions indicated occupancy count > 0. The longest consecutive count error (Max time without a count = 2) was just 3.99 seconds, while there was never an incorrect occupancy prediction.

TABLE 3. Count and occupancy detection performance in the study pod.

Metric	0 People	1 Person	2 People
Exact-Count Accuracy(%)	92.02%	77.12%	68.99%
Occupancy Detection Error Rate(%)	7.97%	4.87%	0%
Longest consecutive count error (s)	1.05	28.21	3.99
Longest consecutive occupancy error (s)	1.05	28.21	0

Data collected over 10-minute intervals for each occupancy level.

Despite the exact-count accuracy variations, the system consistently demonstrated its ability to detect human presence. The short consecutive count error periods across all tests suggest that while continuous perfect counting can be challenging, the system quickly recovers or correctly identifies occupancy states most of the time. This performance, achieved with on-device processing indicates the practical viability of this low-resolution thermal sensor setup for occupancy monitoring in small, controlled environments like study pods.

V. PERFORMANCE OPTIMIZATION

The experimental validation demonstrates that the proposed deterministic algorithmic pipeline achieves robust real-time occupancy monitoring while maintaining complete privacy through edge-based processing, even on a simple microcontroller like an ESP32. Each algorithmic stage contributes critically to this success. The algorithm's performance is governed by several tunable parameters, empirically optimized through testing, as detailed in Table 4. The particular optimal values mentioned indicate the values found out for, and used by the sensor unit during baseline

characterization (Section IV-B1) and tests in the study pod (Section IV-B5). These parameters control the sensitivity and robustness of detection, tracking association, and static object differentiation, with recommended ranges providing flexibility for deployment-specific optimization.

TABLE 4. Algorithm parameters and optimization ranges.

Parameter	Optimal Value	Tuning Range
$T_{\text{threshold}}$	$T_{\text{DHT}} + 0.1\text{ }^{\circ}\text{C}$	$T_{\text{DHT}} \pm 0.5\text{ }^{\circ}\text{C}$
$\alpha_{dt}, \beta_{conv}$	3, 6 pixels	1-8, 2-12 pixels
τ_c	25 pixels	15-35 pixels
τ_d	5 pixels	3-8 pixels
τ_J	0.25	0.2-0.4
N_{persist}	100 frames	80-150 frames
Δ_{static}	0.5 pixels	0.2-1.0 pixels
Δ_{reset}	2.0 pixels	$\geq \Delta_{\text{static}}$

Tuning different parameters has an effect on the system output in specific ways. Dynamic threshold offset $T_{\text{threshold}}$ requires reduction when the thermal contrast between human signatures and DHT temperature obtained is low. Central point thresholds α_{dt} and β_{conv} inversely control detection sensitivity—higher values yield fewer, more selective central points but risk missing irregular blob structures. Tracking association parameters τ_c , τ_d , and τ_J determine re-identification robustness; higher values promote consistent tracking of the same individual across frames, but excessively high values may incorrectly associate new blobs with old ones when transitions occur. Conversely, very low values may cause repeated re-identification of the same individual, leading to miscounts. Static object detection parameters N_{persist} , Δ_{static} , and Δ_{reset} define the boundary between human micro-movements and static objects. N_{persist} and Δ_{static} should be increased if human subjects are misclassified as static, while Δ_{reset} should be raised if static objects are too easily reclassified as dynamic due to subtle heat signature fluctuations.

During the setup of the sensor unit in a specific room or area, parameter values can be determined empirically or from prior experience and fixed in the script for optimal occupancy detection and people counting. These settings remain effective unless there are drastic changes in weather or environmental conditions.

VI. CONCLUSION

The testing across the 5 scenarios revealed that the performance and accuracy were not uniform, as they depended strongly on the thermal contrast characteristics of the surroundings of the scene. The scenarios were chosen to represent a spectrum of possible realistic deployment conditions—ranging from typical indoor spaces to more challenging outdoor spaces—thereby highlighting both limitations and opportunities for practical use. The first scenario thoroughly characterized the operational limits of

a wall-mounted setup by quantifying the effective detection volume and the maximum occupant capacity at each location within that volume (establishing an upper limit of up to four occupants, at a distance of 3 m away from the sensor, with a 53.1° viewing angle). It also demonstrated that during frontal approach, blob sizes increase while centroid shifts remain sufficiently small for reliable tracking, yet large enough to avoid persistence-based masking. In the second scenario, a classroom, the system achieved $\geq 94\%$ accuracy in reporting occupancy status (≥ 1 when people were present within the FoV, and 0 otherwise). The third scenario explored more challenging conditions, where accuracy was 60% for masking static objects as non-humans, and 80% in correctly identifying human counts. The fourth scenario focused on assessing the sensor's practical range; a long room with the sensor obliquely wall-mounted showed a maximum detection distance of 2.625 m. Finally, the fifth scenario—a small two-seater library pod—reflected the most favorable deployment case, achieving $\leq 8\%$ error for occupancy detection.

Although scenarios involving human subjects remaining stationary for extended periods—such as during sleep—may present a challenge by potentially triggering the static object differentiation mechanism, the proposed system is designed to maintain stability through two primary safeguards. First, the algorithm's performance is governed by tunable parameters, which can be empirically optimized for specific deployment environments. Second, physiological studies demonstrate that even in highly “still” states like deep sleep, humans exhibit detectable thermal micro-dynamics. These include involuntary respiration-related chest movements and vasodilation-driven skin temperature fluctuations, both of which generate sufficient spatial-temporal jitter in the blob centroid to prevent incorrect masking [40], [41]. However, in extreme cases where thermal signatures might remain exceptionally static, or in instances of diminished thermal contrast and varying environmental conditions, the hot-object differentiation feature may reach its operational threshold, and can be suspended so that the system prioritizes continuous identification and tracking of human thermal blobs, supporting a more consistent occupancy status.

To contextualize our system's performance, Table 5 presents a comparative analysis against existing methods for human detection using low-resolution thermal sensors. The best-case accuracy scores and inference times are considered in each case, for a 1-person scenario. It is important to note that while this table provides a valuable overview, a direct comparison is challenging due to the inherent differences in evaluation procedures, like our exact-count accuracy vs. mAP metric, hardware platform used, sensor configurations, detection distances, and scene conditions (indoor/outdoor). The analysis reveals that deep learning-based object detection frameworks, such as YOLOv5, achieve exceptional detection accuracy ($>98\%$ mAP) but are critically dependent on powerful, energy-intensive GPU hardware for real-time inference. Our work provides a

more comprehensive evaluation by measuring count and occupancy accuracies separately, over continuous 10-minute periods, achieving a maximum consecutive count error of only 28 seconds. This temporal stability demonstrates near-perfect reliability for real-world deployment: any connected decision-making system operating with a 30-second decision window will achieve correct count-based decisions almost all the time, as brief count variations lasting only a few seconds do not influence the overall system performance.

These results demonstrate that our novel lightweight, deterministic algorithm successfully bridges the gap between high accuracy and the stringent constraints of deeply embedded, privacy-preserving sensing applications, establishing its suitability for real-world deployment.

TABLE 5. Comparative analysis of human-detection systems using low-resolution thermal sensors.

System / Task	Accuracy (People Counting)	Inference Time	Hardware
YOLOv5 [28]	99.5% (mAP@0.5)	1.3 ms	NVIDIA P4000 GPU
YOLOv6 [29]	98.0% (AP@50)	14.8 ms	NVIDIA RTX 3090 GPU
Super-Resolution [31]	83.4% (mAP@50)	5400 ms	AMD RX570 GPU
Lightweight YOLOv5 [30]	98.4% (mAP@0.5)	Not reported Quantitatively	NVIDIA RTX 3080 GPU
Image Processing [33]	95%	30 ms	Standard PC (CPU)
Algorithmic Approach [9]	Not reported Quantitatively	90 ms	Raspberry Pi
This Work	77.12%	133ms	ESP32 MCU

The real-world applicability of the proposed system lies in its ability to deliver privacy-preserving occupancy detection and people counting without reliance on cameras or invasive sensing. By providing both a Boolean occupancy status and precise count values, the framework enables direct integration into smart building controls—for example, automatically switching off lighting, Heating, Ventilation, and Air Conditioning (HVAC), or other appliances when rooms are vacant. This can also support safety and security applications, including ensuring occupancy limits are not exceeded or triggering alerts in emergency evacuations. Since the data is available directly on the sensor and can be accessed through any communication port, the system can be readily scaled into larger networks of IoT devices.

VII. LIMITATIONS AND FUTURE WORK

While the proposed system successfully demonstrates a robust, on-device solution, its design choices and the inherent nature of the sensor modality present several limitations and opportunities for future research.

A primary challenge stems from the fundamental limits of the AMG8833 sensor. Although the math-based algorithm is highly efficient, its ability to resolve individuals in extremely close proximity is constrained, leading to blobs merging

and miscounting. Furthermore, the system's performance is inherently tied to the thermal contrast between temperatures obtained from the subject and their environment. In conditions where the DHT temperature approaches that of the blob region, or in environments with significant thermal noise like direct sunlight, the system's detection capability diminishes. To overcome this fundamental limitation, future work should investigate more advanced generations of thermal sensor arrays with enhanced sensitivity and improved noise rejection.

The persistence-based logic for static heat source rejection, while effective in the tested scenarios, showed sensitivity in dynamic outdoor environments. A logical next step is to evolve this mechanism from a fixed-threshold system to a dynamically adaptive one. Future iterations could adjust the persistence and movement thresholds based on real-time environmental metrics, and past data such as the variance in ambient temperature, and recent fluctuations to better distinguish deliberate human motion from environmental thermal fluctuations. Furthermore, advanced testing methods need to be developed to enable statistical modeling of this feature, leading to more accurate predictions of its performance in the long-term.

To ensure complete practical viability, any future research will also require more extensive, long-term deployments to validate the system's reliability and generalizability across a wider range of participants and challenging environmental conditions.

CODE AND DATA AVAILABILITY

The source code and the dataset used in this study have been made publicly available. The GitHub repository, available at <https://github.com/kernelops/thermal-occupancy-monitor>, contains the complete source code, including the embedded C/C++ implementation for the ESP32 microcontroller and the Python script used for server-side processing and visualization.

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SUHAS PAPANASHI (Member, IEEE) is currently pursuing the degree in electronics and telecommunication engineering with the RV College of Engineering, Bengaluru, India. His research work includes a published international conference paper on road traffic management using computer vision. He has gained research and industry experience through internships at the Samsung Research and Development Institute (PRISM Program), focusing on advanced image processing, and his college's Centre of Excellence in Computer Vision.

He is a member of the IEEE Computer Society and IETE, with experience organizing hackathons and technical events.



NISHANT VASANTKUMAR HEGDE (Member, IEEE) is currently pursuing the degree with the Department of Computer Science and Engineering, RV College of Engineering, Bengaluru, India. He has worked on several related projects and interned at the Samsung Research and Development Institute under the Samsung PRISM Program, contributing to advanced research in the field of image processing. His research interests include AI, ML, cybersecurity, and cryptography.

He is serving on the IEEE Executive Committee of IEEE-RVCE. He has won multiple IEEE-organized hackathons and published several research papers at international IEEE conferences, including one at IIT Indore on a novel encryption algorithm for resource-constrained embedded systems.



MANOJITH BHAT V. is currently pursuing the Bachelor of Engineering degree in information science and engineering with the RV College of Engineering, Bengaluru. He has worked on projects related to machine learning, cybersecurity, the Internet of Things (IoT), and intelligent backend systems. This marks his first publication, where he presents a study on human presence detection using the grid-EYE thermal sensor, focusing on its potential applications. His research interests include machine learning, data science, scalable system architectures, embedded IoT applications, and secure data-driven infrastructures.



R. BHAGYA (Senior Member, IEEE) received the M.E. and Ph.D. degrees. She is currently an Associate Professor with the Department of Electronics and Telecommunication, RV College of Engineering, Bengaluru, with 16 years of teaching experience. She has authored more than 50 publications across international journals and conferences. She recently received the U.K. patent grant for “Battery Thermal Management Solutions for Electric Vehicles” and is a Co-PI on a consultancy project for developing an “Autonomous Agribot.” She has successfully guided 31 undergraduate and 29 postgraduate projects. Her research interests include wireless communication, networks, autonomous vehicles, V2V communication, and 5G.



SATISH PATEL (Senior Member, IEEE) is currently the Founder of DyuLabs Pvt. Ltd., and an Expert of embedded systems. Before founding his company, he built an extensive career holding key engineering roles at major technology companies, including a Lead Engineer at Texas Instruments and an Android Engineer at Linaro. His work focuses on Linux/Android internals, system optimization, and HALs. His contributions include advancements for Google’s Project Ara and the TI Sitara Android Devkit.

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B. ROJA REDDY (Senior Member, IEEE) received the M.Tech. and Ph.D. degrees. She is currently an Associate Professor with the Department of Electronics and Telecommunication, RV College of Engineering, Bengaluru, with 18 years of teaching experience. She has published 36 papers across journals and conferences and has authored one book chapter. She has a strong record of mentorship, having guided ten undergraduate and 30 postgraduate projects, as well as one Ph.D. student. Her research interests include MIMO radars and signal processing. She is an IETE Fellow.